Semester Project List (2026)

Deep Learning — Dr. Zohair Ahmed

ID	Project Title	Domain	Three Representative Papers
1	Implicit Neural Representations & Continuous Networks	Implicit Neural Reps	1) Sitzmann et al., "Implicit Neural Representations with Periodic Activations (SIREN)", NeurIPS 2020 2) Dupont et al., "Coin: Compression with Implicit Neural Representations", ICLR 2022 3) Mildenhall et al., "NeRF: Neural Radiance Fields", ECCV 2020
2	Self-Supervised Learning for Vision & Beyond	Self-Supervised Learning	1) Grill et al., "BYOL", NeurIPS 2020 2) Chen et al., "SimCLR", ICML 2020 3) Caron et al., "DINOv2", CVPR 2024
3	Physics-Informed Neural Networks (PINNs)	Physics-Aware DL	1) Raissi et al., "Physics-Informed Neural Networks", JCP 2020 2) Cai et al., "Deep Learning of PDEs with PINNs", SIAM 2021 3) Wang et al., "Understanding PINN Failure Modes", NeurIPS 2023
4	Neural Architecture Search (NAS) & AutoML	NAS/AutoML	1) Zoph & Le, "Neural Architecture Search with Reinforcement Learning", ICLR 2017 2) Pham et al., "Efficient NAS", ICLR 2018 3) Liu et al., "Once-for-All Networks", ICLR 2020
5	Deep Generative Models (VAEs / GANs / Flows)	Generative Models	1) Kingma & Welling, "Auto-Encoding Variational Bayes", ICLR 2014 2) Goodfellow et al., "GANs", NeurIPS 2014 3) Kingma et al., "Glow: Flow-based Models", NeurIPS 2018

6	Graph Neural Networks (GNNs)	Graph DL	1) Kipf & Welling, "GCNs", ICLR 2017 2) Velickovic et al., "Graph Attention Networks", ICLR 2018 3) Dwivedi et al., "Benchmarking GNNs", ICLR 2022
7	Neural ODEs & Continuous-Depth Models	Neural ODEs	1) Chen et al., "Neural ODEs", NeurIPS 2018 2) Kidger et al., "Neural Controlled Differential Equations", NeurIPS 2020 3) Dupont et al., "Augmented Neural ODEs", NeurIPS 2019
8	Optimization & Implicit Bias of Deep Learning	Optimization Theory	1) Soudry et al., "Implicit Bias of SGD on Linear Networks", ICML 2018 2) Gunasekar et al., "Characterizing Implicit Bias", NeurIPS 2018 3) Arora et al., "Optimization Landscapes", NeurIPS 2019
9	Adversarial Robustness & Certification	Robustness	1) Madry et al., "Adversarial Training", ICLR 2018 2) Wong & Kolter, "Provable Robustness", ICML 2018 3) Salman et al., "Certified Robustness via Randomized Smoothing", ICML 2019
10	Meta-Learning & Few-Shot Learning	Meta-Learning	1) Finn et al., "MAML", ICML 2017 2) Snell et al., "Prototypical Networks", NeurIPS 2017 3) Rusu et al., "Meta-Learning Shared Hierarchies", ICLR 2018
11	Neural Rendering & Vision Synthesis	Neural Rendering	1) Mildenhall et al., "NeRF", ECCV 2020 2) Zhang et al., "NeRF in the Wild", CVPR 2022 3) Yu et al., "PlenOctrees", ICLR 2022
12	Differentiable Programming & Soft Robotics	Differentiable Programming	1) Degrave et al., "DiffTaichi: Differentiable Programming", NeurIPS 2021 2) Li et al., "Differentiable Physics Engines", ICLR 2020 3) Leutenegger et al., "Differentiable SLAM", ECCV 2020
13	Deep Reinforcement Learning	Deep RL	1) Mnih et al., "DQN", NeurIPS 2015

			2) Schulman et al., "PPO", ICML 2017 3) Haarnoja et al., "Soft Actor-Critic", ICLR 2018
14	Transformers Beyond NLP (Vision / Time Series)	Vision Transformers	1) Dosovitskiy et al., "ViT", ICLR 2021 2) Zhou et al., "TimeSformer", ICCV 2021 3) Zheng et al., "STAM: Spatial-Temporal Attention", CVPR 2023
15	Causal Representation Learning	Causal DL	1) Pearl & Bareinboim, "Causal Inference for ML" [Survey] 2) Schölkopf et al., "Causal Representation Learning", ICLR 2021 3) Von Kügelgen et al., "CausalVAE", NeurIPS 2020
16	Energy-Efficient Deep Learning & Model Compression	Efficient DL	1) Han et al., "Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding", ICLR 2016 2) Frankle & Carbin, "The Lottery Ticket Hypothesis", ICLR 2019 3) Lin et al., "HAQ: Hardware-Aware Automated Quantization", CVPR 2019
17	Continual Learning & Catastrophic Forgetting	Lifelong Learning	1) Kirkpatrick et al., "Overcoming Catastrophic Forgetting in Neural Networks", PNAS 2017 2) Aljundi et al., "Memory Aware Synapses", ECCV 2018 3) Farquhar & Gal, "Uncertainty in Continual Learning", ICML 2018
18	Federated & Privacy-Preserving Deep Learning	Privacy & Federated Learning	1) McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data", AISTATS 2017 2) Bonawitz et al., "Practical Secure Aggregation for Federated Learning", CCS 2017 3) Kairouz et al., "Advances and Open Problems in Federated Learning", Foundations & Trends 2021

19	Deep Learning for Spiking Neural Networks & Neuromorphic Computing	Neuromorphic DL	1) Bellec et al., "Long Short-Term Memory and Learning in Spiking Neural Networks", NeurIPS 2018 2) Rueckauer et al., "Conversion of Continuous-Valued Deep Networks to Spiking Neural Networks for Low-Power Applications", IJCNN 2017 3) Sengupta et al., "Going Deeper in Spiking Neural Networks: VGG and Residual Architectures", Front. Neurosci 2019
20	Neural Implicit Control & Differentiable Model Predictive Control	Control & Robotics	1) Amos et al., "Differentiable MPC for End-to-End Learning", NeurIPS 2018 2) Blondel et al., "Efficient Differentiable MPC for Robotics", ICRA 2021 3) Okada et al., "Neural MPC: Learning Dynamics and Constraints for Control", IROS